Chapter 8, Lecture 3

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Some general concepts

Let $X_1, \ldots, X_n$ be iid $f(x|\theta)$ with θ unknown, to be estimated.

An *estimator* $\widehat{\theta}$ of the parameter θ is a function of the data,

$$\widehat{\theta} = \widehat{\theta}(X_1, \ldots, X_n),$$

and therefore random.

If $\mathbb{E}_\theta[\widehat{\theta}^2] < \infty$, we define its *mean squared error* as

$$\mathrm{MSE}(\widehat{\theta}) = \mathbb{E}_\theta[\widehat{\theta} - \theta]^2 = (\mathbb{E}_\theta[\widehat{\theta}] - \theta)^2 + \mathrm{Var}_\theta(\widehat{\theta}).$$

We call $\widehat{\theta}$ *unbiased* if

$$\mathbb{E}_\theta(\widehat{\theta}) = \theta \text{ for all } \theta.$$

An estimator $\widehat{\theta}$ is a *consistent* estimator of θ, or, equivalently we say $\widehat{\theta}$ converges in probability to θ and write

$$\widehat{\theta} \xrightarrow{P} \theta,$$

if

$$\lim_{n\to\infty} \mathbb{P}_\theta\{|\widehat{\theta} - \theta| < \varepsilon\} = 1$$

for all $\varepsilon > 0$. This is the case when

$$\lim_{n\to\infty} \mathrm{MSE}(\widehat{\theta}) = 0.$$

[Note that this is a sufficient condition only.] This is a consequence of the Bienaymé-Chebyshev inequality: For all $\varepsilon > 0$, we have

$$\mathbb{P}_\theta\{|\widehat{\theta} - \theta| \geq \varepsilon\} \leq \frac{\mathrm{MSE}(\widehat{\theta})}{\varepsilon^2}.$$

The *continuous mapping theorem* states that if $T_n = T(X_1, \ldots, X_n)$ converges in probability to a constant μ, and g is a continuous function, then $g(T_n)$ converges in probability to $g(\mu)$.

For instance, let $X_1, \ldots, X_n$ be independent with $\mathbb{E}|X| < \infty$. The law of large numbers states that $\overline{X}_n := \sum_{i=1}^n X_i/n$ converges in probability to $\mu := \mathbb{E}[X]$. Hence, by the continuous mapping theorem, $1/\overline{X}_n$ converges in probability to $1/\mu$, provided $\mu \neq 0$.

The *delta-method* states that for $X_1, \ldots, X_n$ independent, identically distributed with mean $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}[X_i] = \sigma^2$, we have for any g that is continuously differentiable at μ,

$$\lim_{n\to\infty} \sup_{t\in\mathbb{R}} \left| \mathbb{P}\left\{ \sqrt{n}\frac{g(\overline{X}_n) - g(\mu)}{|g'(\mu)|\sigma} \leq t \right\} - \Phi(t) \right| = 0.$$

In words, we say that the distribution of

$$g(\overline{X}_n) \text{ is approximately normal } N(g(\mu), [g'(\mu)]^2\sigma^2/n) \text{ for large } n.$$

The reason why can be seen as follows: By a Taylor expansion, we have

$$\widehat{\theta} = g(\overline{X}_n) \quad = \quad g(\mu) + g'(\mu)(\overline{X}_n - \mu) + \cdots$$

and the term on the right, ignoring the remainder terms (the dots),

$$g(\mu) + g'(\mu)(\overline{X}_n - \mu)$$

is asymptotically normal by the CLT, with mean

$$\mathbb{E}[g(\mu) + g'(\mu)(\overline{X}_n - \mu)] = g(\mu)$$

and variance

$$\begin{aligned}\mathrm{Var}\left(g(\mu)+g'(\mu)(\overline{X}_n-\mu)\right) &= \mathrm{Var}\left(g'(\mu)\overline{X}_n\right)\\ &= \{g'(\mu)\}^2\mathrm{Var}\left(\overline{X}_n\right)\\ &= \{g'(\mu)\}^2\frac{\sigma^2}{n}\end{aligned}$$

We continue with our example. We have independent observations $X_1,\ldots,X_n$ from an Exponential distribution with unknown parameter $\lambda>0$. We will use that $\mathbb{E}[X]=1/\lambda$ and $\mathrm{Var}(X)=1/\lambda^2$. In addition, recall that

$$\mathbb{E}[\overline{X}_n]=\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n X_i\right]=\frac{1}{n}\sum_{i=1}^n\mathbb{E}\left[X_i\right]=\mathbb{E}[X_1]=\frac{1}{\lambda}$$

and

$$\begin{aligned}\mathrm{Var}(\overline{X}_n) &= \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^n X_i\right)\\ &= \frac{1}{n^2}\sum_{i=1}^n\sum_{j=1}^n\mathrm{Cov}(X_i,X_j)\\ &= \frac{1}{n^2}\sum_{i=1}^n\mathrm{Var}(X_i) \qquad \text{since } X_i \text{ are independent}\\ &= \frac{\mathrm{Var}(X_1)}{n}=\frac{1}{n\lambda^2}.\end{aligned}$$

While we have the exact distribution of

$$\widehat{\lambda}=\frac{1}{\overline{X}_n}$$

and a closed form of its density could be derived from the fact that this distribution is n divided by a Gamma distribution with parameters n and λ, we are not always in the situation that we have a closed form, and use an approximation based on a simple Taylor expansion. Let $g(t)=1/t$ so that $g'(t)=-1/t^2$. Since $\overline{X}_n$ is close to

$\mu = \mathbb{E}[X] = 1/\lambda$, in probability, as $n \to \infty$ by the LLN, we can write

$$\begin{aligned} \widehat{\lambda} := g(\overline{X}_n) &= g(\mu) + (\overline{X}_n - \mu)g'(\mu) + \cdots \\ &= \lambda - \lambda^2(\overline{X}_n - \mu) + \cdots \end{aligned}$$

Ignoring the dots (remainder terms), we see that $\widehat{\lambda}$ is essentially

$$\lambda - \lambda^2(\overline{X}_n - \mu)$$

This term has mean

$$\mathbb{E}[\lambda - \lambda^2(\overline{X}_n - \mu)] = \lambda$$

and variance

$$\begin{aligned} \mathrm{Var}(\lambda - \lambda^2(\overline{X}_n - \mu)) &= \mathrm{Var}(\lambda^2 \overline{X}_n) \\ &= \lambda^4 \mathrm{Var}(\overline{X}_n) \\ &= \lambda^4 \frac{\mathrm{Var}(X)}{n} \\ &= \frac{\lambda^2}{n} \end{aligned}$$

Compare these approximate mean and vaiance with the exact mean $\lambda n/(n-1)$ and exact variance $\lambda^2 n^2/\{(n-2)(n-1)^2\}$.

By the Central limit theorem (CLT), we know that $\overline{X}_n$, has an approximate normal distribution. Specifically,

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}\left\{ \frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \leq x \right\} - \Phi(x) \right| \to 0, \qquad \text{as } n \to \infty.$$

Here

$$\mu := \mathbb{E}[X], \qquad \sigma^2 := \mathrm{Var}(X), \qquad \Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}t^2)\, \mathrm{d}t.$$

Back to our example, we see that $\widehat{\lambda}$ has an approximate normal distribution with men λ and variance λ^2/n, for large n. This approximation is verified in our small simulation study in R.